

Reviewing index operations

What to do

1 Simplify the following.

a $2^3 \times 2^4 =$

b $3^2 \times 3^3 \times 3^2 =$

c $5^6 \times 5^5 \times 5^3 =$

d $(-6^1) \times (-6^2) =$

e $3^2 \times 5^3 \times 3^4 \times 5^2 =$

f $x^3 \times x^4 \times x^6 =$

2 **MC** The expression $(9a^6b^2) \div (6a^2b^6)$ when simplified is:

A $\frac{3a^3}{2b^3}$

B $\frac{3a^4}{2b^4}$

C $\frac{3a^4b^4}{2}$

D $\frac{3}{2a^4b^4}$

E $\frac{3a^3}{2b^4}$

3 Simplify the following using the first three index laws.

a $x^3y^2 \times x^0y^2$

b $x^7y^5 \div x^{11}y^0$

c $3a^0b^2 \times b^2 \times a^5b^0$

d 0^55^0

e $-6q^0r^2 \div 3qr^0$

f $5k^2q^9 \times 2q^0k$

4 Simplify the following.

a $(2^3)^4$

b $(x^2y^3z^4)^6$

c $(-2k^0p^5)^2$

5 **MC** The result of simplifying $(4^2r^5s^4)^2 \div (4r^4s^0)^3$ is:

A $\frac{s^8}{4r^2}$

B $\frac{4s^7}{r^2}$

C $\frac{4s^8}{r^2}$

D $\frac{16s^8}{r^2}$

E $\frac{64s^8}{r^2}$

6 Simplify the following.

a $\left(\frac{1}{x^2}\right)^3$

b $\left(\frac{y^3}{x^5}\right)^2$

c $\left(\frac{3p^2}{q^3}\right)^4$



Practising the index laws

What to do

1 Simplify the following using the first index law.

a $3^4 \times 3^6 =$

b $4^5 \times 4^3 \times 4^2 =$

c $5^{10} \times 5^8 \times 5^2 =$

d $(-6^5) \times (-6^2) =$

e $3^3 \times 4^3 \times 3^4 \times 4^5 =$

f $x^5 \times x^2 \times x^7 =$

2 **(MC)** The expression $(16a^5b^4) \div (12a^3b^8)$ when simplified is:

A a^2b^2

B $\frac{4a^2}{b^2}$

C $\frac{8a^3}{b^4}$

D $\frac{4a^2}{3b^4}$

E $\frac{4a^2}{b^4}$

3 Simplify the following using the first three index laws.

a $4x^3y^3 \times 4^0x^0y^2$

b $32x^7y^4 \div 12x^{10}y^0$

c $3a^0b^0c^2 \times c^2 \times 6a^5b^3$

d 0^3x^0

e $-9p^3q^0r^2 \div 3p^5q^3r^0$

f $5kq^9 \times 4qk^4m^0$

4 Simplify the following.

a $(4^3)^2$

b $(2x^3y^4z^2)^5$

c $(-3d^1k^0p^6)^3$

5 **(MC)** The result of simplifying $(3^2r^3s^9)^2 \div (3r^4s^{10})^3$ is:

A $\frac{1}{r^6s^{12}}$

B $\frac{3}{r^6s^{12}}$

C $\frac{27}{r^6s^{12}}$

D $\frac{3r^6}{s^{12}}$

E $3r^6s^{12}$

6 Simplify the following.

a $\left(\frac{1}{x^6}\right)^2$

b $\left(\frac{y^2}{x^3}\right)^4$

c $\left(\frac{3p^4}{4q^7}\right)^2$